

Evidence-aligned immunometabolic analysis in human intracerebral hemorrhage

Verified data types, corrected longitudinal unit of analysis, sensitivity tests, and interpretation boundary

a Verified datasets

GSE163256

Longitudinal sorted-cell transcriptomics
CD14+ monocytes/macrophages and neutrophils
21 ICH patients + 5 healthy controls

Locked monocyte analysis subset

136 unique ICH observations
20 patients; days 1–6
technical duplicates collapsed

GSE166638

scRNA-seq from one longitudinally sampled patient
exploratory cellular triangulation only

b Corrected primary models

Patient-level mixed models

Inflammation: $\beta = 0.319$ (95% CI 0.131 to 0.507)
Hypoxia: $\beta = 0.831$ (95% CI 0.690 to 0.972)

outcome ~ contrast + compartment + day + (1 | patient)

Overlap-pruned sensitivity

Inflammation: $\beta = 0.372$ (95% CI 0.144 to 0.601)
Hypoxia: $\beta = 0.647$ (95% CI 0.450 to 0.845)

All cross-signature overlaps removed

c Interpretation boundary

Supported

Expression-derived metabolic scores
co-vary with hypoxia and inflammation
in longitudinal sorted monocyte profiles

Not established

Metabolic flux or causal niche effects
population-level scRNA-seq replication
or therapeutic actionability